Power from batteries and the sun



IoT · Level 3 · Lesson 4

Today you will

- Explain why a device's radio uses more battery than anything else.
- Make a battery last longer by sleeping and sending less often.
- Describe how a small solar panel keeps a sensor running.

Make it last

- Hours = capacity ÷ average current
- The radio is the hungriest part
- Sleep most of the time; send less
- Sun by day, battery by night

Battery sums

1. Find two or three mAh labels.
2. Plan with 70% of each label.
3. Work out the hours at 100 mA and at 10 mA.
4. Plan a garden sensor: what, how often, which radio.
5. Decide: does it need a solar panel?

Never open, crush, puncture or heat a battery, and never leave lithium batteries or power banks in hot sun. Keep button batteries away from small children.

Let’s discuss

1. Which data around you must be fresh every second, and which could wait an hour?
2. Where at your school would a solar-powered sensor be useful?
3. Why does a coin cell suit Bluetooth LE but not Wi-Fi?

Check yourself

- A sensor sends every minute over Wi-Fi. Name two ways to make its battery last longer.